

Chidaksh Ravuru

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Research Interests

My research investigates why NLP models fail in predictable ways, using causal methods to discover and verify spurious shortcuts that classifiers learn to exploit. I am broadly interested in building interpretability tools that generate natural language explanations of model behavior, designing evaluation frameworks that catch systematic errors before deployment, and applying these methods to improve the reliability of enterprise RAG and agentic systems.

Education

- 2024-Present **University of North Carolina at Chapel Hill**, MS in Computer Science
Advisor: Prof. Shashank Srivastava, GPA: 4.0/4.0
- 2020-2024 **Indian Institute of Technology, Dharwad**, B.Tech in Computer Science
Advisors: Prof. Prabuchandran K.J. & Prof. Rajshekhar Bhat, GPA: 9.24/10

Main Conference Papers

- 2026 **UNMASK: Discovering and Causally Verifying Spurious Shortcuts in Text Classifiers**, COLM, Under Review
Chidaksh Ravuru, Shashank Srivastava
- 2024 **Agentic Retrieval-Augmented Generation for Time Series Analysis**, KDD
Chidaksh Ravuru, Sagar Srinivas Sakhinana, Venkataramana Runkana
- 2026 **Can LLMs Understand What We Cannot Say?**, ACM FAccT
Anika Sharma, Malvika Mampally, **Chidaksh Ravuru**, Kandyce Brennan, Neil Gaikwad

Workshop Papers

- 2024 **Parameter-Efficient Quantized Mixture-of-Experts Meets Vision-Language Instruction Tuning**, ICML, Workshop on Foundational Models
Sagar Srinivas Sakhinana, **Chidaksh Ravuru**, Geethan Sannidhi, Venkataramana Runkana

Research Experience

- Summer 2025 **Applied Scientist Intern, Pavo AI**
Trained persona-conditioned Process Reward Models (PRMs) to guide search-space exploration in a deployed planning agent
- Built **PersonaBench**, a synthetic dataset of **3,000 positive/negative trajectories** using a **student-teacher framework**: student extracts company profiles from the web, teacher verifies sources and flags hallucinations, then LLMs generate persona-conditioned reasoning traces reflecting each company's ML maturity and infrastructure constraints.
 - Trained **persona-conditioned PRMs** on PersonaBench, conditioning on both **correctness and persona fit** (vs. correctness-only in standard PRMs) to select responses suited to each company's capabilities, improving the planning agent's task performance by **10%** in production.
- Summer 2023 **Machine Learning Research Intern, Tata Research Data and Development Centre**
Designed **agentic RAG system** for time series analysis.
- Developed an agentic RAG model for time series, achieving a **12% improvement** in prediction accuracy on complex temporal forecasting tasks.
 - Boosted model performance by **18%** in anomaly detection and **15%** in pattern recognition across 5+ task-specific benchmarks using finetuned SLMs.

Projects

- 2025–2026 **UNMASK: Causal Verification of Spurious Shortcuts in Text Classifiers**, [COLM 2026](#) | *Under Review*
- Built a **4-stage automated pipeline** (discovery, validation, causal verification, debiasing) using causal mediation analysis and TCAVs. Applied to MNLI, surfacing **7 to 8 causally verified spurious features** from 625 candidates, reducing causal reliance by **60%** and improving ANLI by **+4.6%p** and HANS by **+9.2%p**.
 - Applied DFR debiasing on RoBERTa, improving HANS non-entailment accuracy from **38.9%** to **66.6%** (+22.9%p) with only **−1.4%p MNLI-m** drop. Extended pipeline to **RewardBench2** for analyzing spurious patterns in reward model preference data.
- 2024 **Automated Commentary Generation for Soccer Highlights**, [Code](#) | [Report](#)
- Built a **browser automation pipeline** using Playwright and BeautifulSoup to collect video data from YouTube, handling dynamic page rendering, rate limiting, and structured metadata extraction.
 - Fine-tuned a **video-language model** with a progressive fine-tuning strategy after diagnosing a temporal consistency failure mode, improving **BLEU-4 by 322%** and **CIDEr by 72%**.
- 2024 **Intelligent Planner: Dynamic LLM Query Routing System**, [Code](#) | [Report](#)
- Fine-tuned a **Llama-3.1-8B** classifier on a custom synthetic dataset to route queries to specialized LLMs, improving performance on **MMLU by 5%** and **GSM8K by 3%**.
 - Engineered end-to-end deployment with a **FastAPI** backend, React frontend, and a **continuous evaluation system** that monitors model quality and flags regressions.
- 2024 **Comparative Analysis of LLM Agents for API Tool Use**, [Code](#) | [Report](#)
- Evaluated agentic reasoning strategies (**CoT**, **ReAct**, **DFSDT**) across tool-selection accuracy and response latency to identify the optimal approach for production deployment.
 - Demonstrated that CoT achieved **87.5% tool-selection accuracy** while being **5× faster** than the more complex DFSDT-based ToolLLM framework.

Skills

Courses	Reinforcement Learning, Reliable Machine Learning, Advanced LLM Agents, RLHF/RLAIF, NLP, Causal Inference
Programming	Python, C, C++, JAX, Java, Javascript, Bash/Shell Script, SQL, React, NodeJS, TypeScript, Django, L ^A T _E X
Libraries	PyTorch, HuggingFace Transformers, LangChain, LangGraph, LlamaIndex, TensorFlow, PEFT/LoRA, Pandas, NumPy, Scikit-learn, FAISS
Software	Git, Docker, AWS, Azure, Weights & Biases, PySpark, MySQL, Unix/Linux, Chroma, Pinecone, sentence-transformers, vLLM, Ollama

Teaching & Achievements

- Summer 2025 **Summer School Teacher**, *UNC School of Medicine*
Led a lecture series for high school students on Statistics and Machine Learning (Linear Regression, SVMs) in neuroscience and medical imagery.
- 2025–Present **Graduate Teaching Assistant**, *UNC School of Data Science and Society*
TA for DATA 110: Intro to Data Science. Conducted lab recitations and taught basic probability, statistics, and Python coding for undergraduates.
- Reviewing **ACL, EMNLP, ICLR, COLM**
- 2024 Selected as one of the **top 1%** of applicants countrywide for Google Research Week, India